

This technical report provides the detailed pseudocode of the proposed greedy algorithms and the proofs of the theoretical results. Algorithm 1 (Followers) describes the procedure for computing the follower set induced by an anchored edge. Algorithm 2 (GIA) presents the baseline cost-aware greedy anchoring strategy. Algorithm 3 (GPA) extends GIA by incorporating component-wise pruning and upper-bound analysis, substantially reducing the search space while preserving the same greedy selection criterion. Note that, although the anchored k -core problem is conceptually related to WASCO, the additional flexibility of weighted reinforcement and multi-unit budget allocation prevents a straightforward reduction from the former to the latter. We therefore establish hardness directly via a reduction from the Maximum Coverage problem [2].

A Pseudo code of Followers

Algorithm 1: Followers(G, e, δ_e, s, C_s)

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1 PQ ← queue(); // sort by coreness-layer ascending order
2  $\mathcal{F} \leftarrow \emptyset$ ;
3  $w(e) \leftarrow w(e) + \delta_e$ ;
4 for each  $x \in \{u, v\}$  if  $x \notin C_s$  do
5   PQ.add( $x$ );
6 while PQ  $\neq \emptyset$  do
7    $x \leftarrow \text{PQ.pop}()$ ;
8    $w^+(x) \leftarrow \text{SupportWeightedDegree}(G, x)$ ;
9   if  $w^+(x) \geq s$  then
10     $\mathcal{F} \leftarrow \mathcal{F} \cup \{x\}$ ;
11    for each  $y \in N(x, G)$  and  $x < y$  do
12      if  $y \notin C_s$  and  $y \notin \mathcal{F}$  then
13        PQ.add( $y$ );
14   else
15     Q ← queue();
16     Q.push( $x$ );
17     while Q  $\neq \emptyset$  do
18        $y \leftarrow \text{Q.pop}()$ ;
19       if  $y \in \mathcal{F}$  then
20          $\mathcal{F} \leftarrow \mathcal{F} \setminus \{y\}$ ;
21       for each  $z \in N(y, G) \cap \mathcal{F}$  do
22          $w^+(z) \leftarrow w^+(z) - w(y, z)$ ;
23         if  $w^+(z) < s$  then
24           Q.push( $z$ );
25  $w(e) \leftarrow w(e) - \delta_e$ ;
26 return  $\mathcal{F}$ 

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Algorithm description. Algorithm 1 computes the follower set induced by an anchored edge via a localised simulation of the s -core peeling process. Starting from the anchored endpoints, the algorithm explores only nodes that are reachable through neighbours with higher coreness-layer pairs, following the upstairs order established by the onion-layer structure. A node is retained as a follower if its support weighted degree meets the s -core threshold; otherwise, it is discarded. Discarding a node may reduce the support of neighbouring retained followers, potentially triggering further removals. This cascading update exactly mirrors the behaviour of the s -core decomposition, but is restricted to the structurally affected region. The process terminates when no further admissible nodes remain, and the surviving candidates constitute the final follower set.

Algorithm 2: GIA(G, b, s)

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1  $G' \leftarrow G, A \leftarrow \emptyset, \text{sum} \leftarrow 0$ ;
2  $C_s \leftarrow s\text{-core}(G')$ ;
3 while  $\text{sum} < b$  do
4    $E^* \leftarrow \binom{V}{2} \setminus \binom{C_s}{2}$ ;
5    $e^* \leftarrow \text{none}, \delta^* \leftarrow 0, \text{max\_FR} \leftarrow 0$ ;
6   for each  $e \in E^*$  do
7      $\delta_e \leftarrow \text{computeDelta}(G', s, e)$ ;
8     if  $\text{sum} + \delta_e \leq b$  then
9        $\mathcal{F}(e, \delta_e) \leftarrow \text{Followers}(G', e, \delta_e, s, C_s)$ ;
10       $\text{FR}(e) \leftarrow |\mathcal{F}(e, \delta_e)| / \delta_e$ ;
11      if  $\text{max\_FR} < \text{FR}(e)$  then
12         $e^* \leftarrow e, \delta^* \leftarrow \delta_e$ ;
13         $\text{max\_FR} \leftarrow \text{FR}(e)$ ;
14    $w_{G'}(e^*) \leftarrow w_{G'}(e^*) + \delta^*$ ;
15    $\text{sum} \leftarrow \text{sum} + \delta^*$ ;
16    $A \leftarrow A \cup (e^*, \delta^*)$ ;
17    $C_s \leftarrow s\text{-core}(G')$ ;
18 return  $A$ 

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B Pseudo code of GIA

Algorithm description. Algorithm 2 presents the baseline greedy incremental anchoring procedure of GIA. Starting from the initial s -core, the algorithm iteratively evaluates candidate edges whose endpoints are not both contained in the current s -core. For each candidate edge, GIA determines the required weight increment using the MEE rule, computes the induced follower set, and evaluates the corresponding follower ratio. At each iteration, the edge achieving the largest follower ratio is selected, its weight is increased accordingly, and the s -core is updated. This process repeats until the available budget is exhausted, yielding a greedy baseline for comparison.

C Pseudo code of GPA

Algorithm description. Algorithm 3 presents the pseudocode of GPA, an accelerated variant of GIA that reduces the candidate search space while preserving the same greedy selection criterion. GPA exploits the locality of follower propagation by organising nodes outside the s -core into connected components and maintaining, for each component, the best edge-weight increment candidate pair identified so far. At each iteration, GPA selects the current best candidate across all components. It then explores candidate edges whose endpoints belong to different components, guided by node-level and edge-level upper bounds that allow early termination and safe pruning. Only candidate pairs that cannot be ruled out by these bounds are evaluated using the same follower computation as in GIA. After applying the selected edge reinforcement, GPA updates the component structure. If the reinforced edge connects two distinct components, they are merged and their best candidate is recomputed; otherwise, only the affected component is updated. In both cases, the s -core is recomputed locally within the affected components. This process repeats until the budget is exhausted, yielding the same greedy outcome as GIA with fewer candidate evaluations.

Algorithm 3: GPA(G, b, s)

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1  $G' \leftarrow G, A \leftarrow \emptyset, \text{sum} \leftarrow 0;$ 
2  $L \leftarrow \text{list}(), M \leftarrow \text{map}();$  // maps each component to its
   best  $(e, \delta_e, FR)$ 
3  $C_s \leftarrow s\text{-core}(G');$ 
4  $M \leftarrow \text{buildComponents}(G', b, s);$ 
5 while  $\text{sum} < b$  do
6    $e^*, \delta^*, \text{max\_FR} \leftarrow \arg \max_{(e, \delta_e, FR) \in M} FR;$ 
7   for  $u \in V \setminus C_s$  and  $\text{sum} + (s - c(u)) \leq b$  do
8      $U(u) \leftarrow \text{UpperBound}(G', u, s);$ 
9      $L.append(u);$ 
10   $\text{sort}(L);$  // by upper bound decreasing order
11  for  $u \in L$  do
12    if  $\text{max\_FR} > 2 \cdot U(u)$  then
13      break;
14    for  $v \in L$  s.t.  $v$  appears after  $u$  do
15      if  $u.\text{component} = v.\text{component}$  then
16        continue;
17      if  $\text{max\_FR} > U(u) + U(v)$  then
18        break;
19      if  $\text{max\_FR} > U(u, v)$  then
20        continue;
21      else
22         $e \leftarrow (u, v);$ 
23         $\delta_e \leftarrow \text{computeDelta}(G', s, e);$ 
24        if  $\text{sum} + \delta_e \leq b$  then
25           $\mathcal{F}(e, \delta_e) \leftarrow \text{Followers}(G', e, \delta_e, s,$ 
26             $C_s);$ 
27           $FR(e) \leftarrow |\mathcal{F}(e, \delta_e)| / \delta_e;$ 
28          if  $\text{max\_FR} < FR(e)$  then
29             $e^* \leftarrow e, \delta^* \leftarrow \delta_e;$ 
30             $\text{max\_FR} \leftarrow FR(e);$ 
31         $w_{G'}(e^*) \leftarrow w_{G'}(e^*) + \delta^*;$ 
32         $\text{sum} \leftarrow \text{sum} + \delta^*;$ 
33         $A \leftarrow A \cup (e^*, \delta^*);$ 
34         $\text{updateComponents}(G', b, s, \text{sum}, M);$ 
35 return  $A$ 

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D Proof of Theorem 1

PROOF. We reduce the Maximum Coverage (MC) problem, which is known to be NP-hard unless $P = NP$ [2], to the WASCO problem. The MC problem is defined as follows: given a collection of sets $\{T_1, T_2, \dots, T_t\}$ containing elements from a universe $\{e_1, e_2, \dots, e_d\}$ and a budget b' , the objective is to select b' sets such that the number of unique elements covered is maximised.

To construct an instance of the WASCO problem, consider an arbitrary instance I_{MC} of the MC problem with sets $\{T_1, \dots, T_t\}$ and elements $\{e_1, \dots, e_d\} = \bigcup_{i=1}^t T_i$, as illustrated in the top of Figure 1. We define an instance $I_{WASCO} = (G, s, b)$ as follows:

- The graph G consists of three components: a part M representing the sets T_i , a part N representing the elements e_j , and a set of $(d+2)$ -cliques.

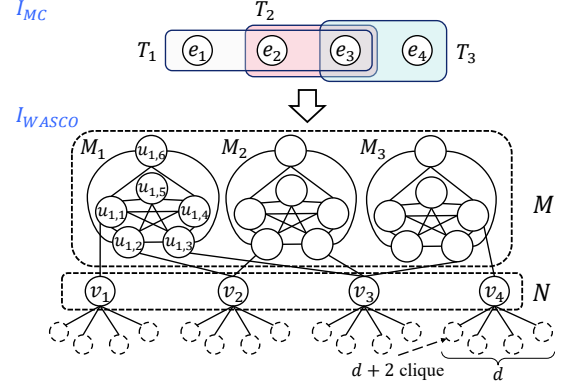


Figure 1: Proof of NP-hardness

- For each set T_i , create a corresponding set of nodes M_i in M , where $M_i = \{u_{i,1}, u_{i,2}, \dots, u_{i,d+2}\}$. The nodes $\{u_{i,1}, \dots, u_{i,d+1}\}$ form a $(d+1)$ -clique, and $u_{i,d+2}$ is connected to $\{u_{i,1}, \dots, u_{i,d}\}$.
- For each element e_j , create a corresponding node v_j in N . Add an edge between v_j and $u_{i,j}$ for every set T_i such that $e_j \in T_i$.
- For each v_j , add d disjoint $(d+2)$ -cliques, and connect v_j to one node in each clique.
- Set $s = d + 1$ and $b = b'$.
- Every edge in G has a weight of 1.

This construction can clearly be completed in polynomial time. Initially, during the s -core decomposition, all nodes in M and N are removed, leaving only the $(d+2)$ -cliques. This happens because:

- Nodes $u_{i,d+1}$ and $u_{i,d+2}$ in M_i have degree d , which is less than $s = d + 1$, and are therefore removed.
- Removing $u_{i,d+1}$ and $u_{i,d+2}$ causes the remaining nodes $u_{i,j}$ in M_i to fail the s -core condition, as their degree drops below s .
- All v_j nodes in N are similarly removed because their connections to M_i are lost.

As a result, only the $(d+2)$ -cliques remain in the s -core after this initial decomposition. To maximise the s -core size, an optimal solution is to add an edge between $u_{i,d+1}$ and $u_{i,d+2}$ in M_i . Adding this edge corresponds to selecting the set T_i in the MC problem. Note that another optimal solution adds edges between distinct sets in M (e.g., $(u_{i,d+2}, u_{j,d+1})$ and $(u_{j,d+2}, u_{i,d+1})$), which can always be replaced by the intra-set connection since both strategies yield the exact same number of followers. When the edge is added:

- All nodes in M_i are included into the s -core because the degree of $u_{i,d+1}$ and $u_{i,d+2}$ becomes $d + 1$.
- All nodes v_j connected to M_i are also included in the s -core, as their degree conditions are satisfied.

Since the optimal way to maximise the s -core is to add edges corresponding to specific T_i sets so as to include as many e_i elements as possible in the s -core, this process directly mirrors the objective of the MC problem to select sets that maximise the coverage of elements. Thus, selecting M_i that are more densely connected to uncovered v_j aligns with the optimisation goals of the MC problem.

Therefore, the MC problem is reducible to the WASCO problem, implying that $I_{MC} \leq_P I_{WASCO}$. As the MC problem is NP-hard, this proves the NP-hardness of the WASCO problem. $\square$

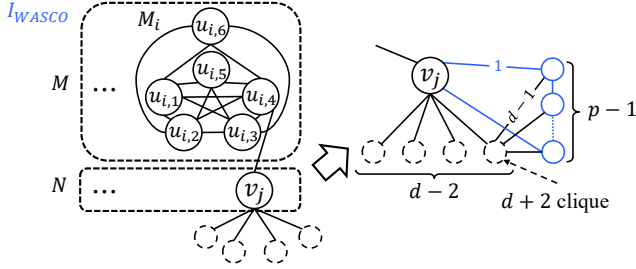


Figure 2: Gadget for the inapproximability reduction

E Proof of Theorem 2

PROOF. We give a gap-preserving reduction from the Maximum Coverage (MC) problem. It is known that, for any $\varepsilon > 0$, MC cannot be approximated within a factor of $(1 - 1/e + \varepsilon)$ in polynomial time, unless $P = NP$ [1].

We start from the construction in the proof of Theorem 1 and apply the same two modifications to each node in part N (Figure 2). For each element node v_j , we (i) reduce the number of disjoint $(d+2)$ -cliques attached to v_j from d to $d-2$, and (ii) replace v_j by a cycle on p nodes by adding $p-1$ new nodes connected to v_j with edges of weight 1. Moreover, each of the $p-1$ new cycle nodes is connected to one node of a (randomly chosen) attached $(d+2)$ -clique by an edge of weight $d-1$. We set $s = d+1$ as in Theorem 1. Let p be an integer chosen as a polynomially bounded function of the MC instance size; its value will be specified at the end of the proof.

Let G be the resulting WASCO instance, and let $f(A) = |C_s(G_A)|$ denote the size of the s -core after applying an anchoring solution A with total budget b . As in Theorem 1, without any anchoring, all nodes in M and N are removed by the s -core decomposition, and only the union of the $(d+2)$ -cliques remains in the s -core. Denote by B the number of nodes in this always-retained clique part; B is fully determined by the MC instance and is independent of A .

Now consider any feasible anchoring solution A that uses exactly b budget units (if it uses fewer, we can add arbitrary weight increments on edges between already-retained clique nodes to exhaust the budget without changing the s -core size). By the same argument as in Theorem 1, anchoring an edge within a set gadget M_i corresponds to selecting the set T_i , which makes all $d+2$ nodes of M_i enter the s -core; since exactly b sets are selected, this contributes an additive term $(d+2)b$ to $f(A)$.

Let $\text{cov}(A)$ be the number of elements covered by the corresponding b selected sets in the MC instance. By construction, $\text{cov}(A)$ is exactly the number of element nodes v_j that enter the s -core under A . We next show that if a node v_j enters the s -core, then all p nodes on the cycle containing v_j also enter the s -core. Indeed, once v_j is retained, every other cycle node has weighted degree $2 + (d-1) = d+1 = s$, coming from its two cycle neighbours (weight 1 each) and its incident edge of weight $d-1$ to a clique node that is always retained. Hence, each cycle node satisfies the s -core condition in the induced subgraph consisting of the always-retained cliques together with this cycle. Therefore, the inclusion of v_j contributes an additional p nodes to the s -core.

Putting the above together, we obtain the following exact relation for every feasible A :

$$f(A) = B + (d+2)b + p \cdot \text{cov}(A).$$

Let OPT_{MC} be the optimal MC value, and let OPT_{W} be the optimal WASCO objective. From the relation above,

$$\text{OPT}_{\text{W}} = B + (d+2)b + p \cdot \text{OPT}_{\text{MC}}.$$

Assume, for contradiction, that there exists a polynomial-time $(1 - 1/e + \varepsilon)$ -approximation algorithm for WASCO. Let $\rho = 1 - 1/e + \varepsilon$, and let A be the solution returned by this algorithm. Then $B + (d+2)b + p \cdot \text{cov}(A) = f(A) \geq \rho \text{OPT}_{\text{W}} = \rho(B + (d+2)b + p \cdot \text{OPT}_{\text{MC}})$, which implies

$$\text{cov}(A) \geq \rho \text{OPT}_{\text{MC}} - \frac{(1-\rho)(B + (d+2)b)}{p}.$$

Choose

$$p \geq \frac{2(B + (d+2)b)}{\varepsilon},$$

which is polynomially bounded in the instance size. Since $\text{OPT}_{\text{MC}} \geq 1$ for any non-trivial MC instance, the additive term is at most $\varepsilon/2$, and thus

$$\text{cov}(A) \geq (1 - 1/e + \varepsilon/2) \text{OPT}_{\text{MC}}.$$

This yields a polynomial-time $(1 - 1/e + \varepsilon/2)$ -approximation for MC, contradicting the inapproximability of MC unless $P = NP$ [2]. Therefore, WASCO cannot be approximated within a factor of $(1 - 1/e + \varepsilon)$ in polynomial time for any $\varepsilon > 0$, unless $P = NP$. $\square$

F Proof of Theorem 3

PROOF. Suppose that f is submodular. Then, for any two sets A and B , it must hold that $f(A) + f(B) \geq f(A \cup B) + f(A \cap B)$. Consider a graph consisting of four nodes $\{u, v, x, y\}$ and two edges (u, x) and (v, y) , each with a weight of 1. Let $s = 2$, if $A = \{(u, v), 1\}$ and $B = \{(x, y), 1\}$, $f(A) + f(B) = 0 < f(A \cup B) + f(A \cap B) = 4$. $\square$

References

- [1] Uriel Feige. 1998. A threshold of $\ln n$ for approximating set cover. *Journal of the ACM (JACM)* 45, 4 (1998), 634–652.
- [2] Samir Khuller, Anna Moss, and Joseph Seffi Naor. 1999. The budgeted maximum coverage problem. *Information processing letters* 70, 1 (1999), 39–45.